MolPallete — Corpus EDA

naveja_recap

/home/mogan/preprocessed/molpallete/coconut-flavordb_full_r033/naveja_recap

generated 2026-08-20 06:46

A · Provenance

field	value
sources	['flavordb', 'coconut']
method	naveja_recap
decomposition_params	{'ratio': 0.3333, 'include_ring': True}
core_ratio	0.3333
max_cores	10
max_rgroups	8
keep_stereo	True
neutralise	False
dedup	True
n_duplicates_removed	4412
size_filter	{'min_heavy_atoms': 5, 'max_heavy_atoms': 50}
graph_hash_version	2
rdkit_version	2026.03.2
created_at	2026-08-20T06:29:01

B · Composition

quantity	value
molecules	413,459
from coconut	391,965 (94.8%)
from flavordb	21,494 (5.2%)
decompositions	3,455,836
per molecule	mean 8.36, median 10, max 10
R-group slots	3,851,767
per decomposition	mean 1.11
k >= 2	9.47% of decompositions

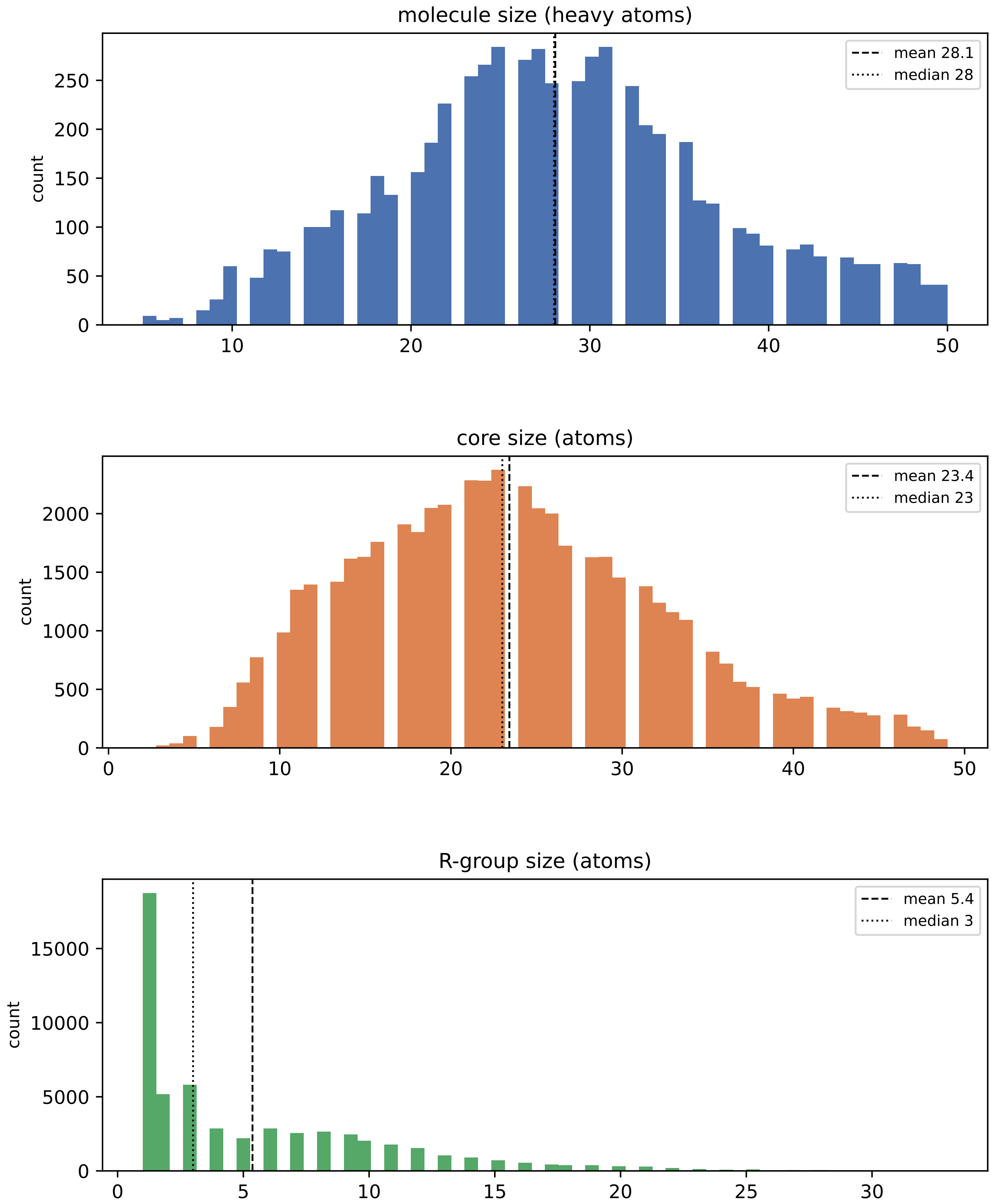
k>=2 is what the islinked subset space and the core-decoration objective have to work with.

C · Augmentation space

scheme	size
sampling_unit = molecule	413,459 instances/epoch
sampling_unit = decomposition	3,455,836 instances/epoch (8.4x)
full (mol, core, islinked) space	4,493,636 (10.9 per molecule)

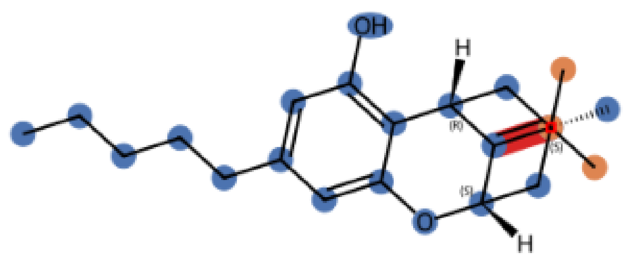
MolPLA's two stages: one molecule yields many cores; one core yields 2^k - 1 islinked subsets.

D · Size distributions

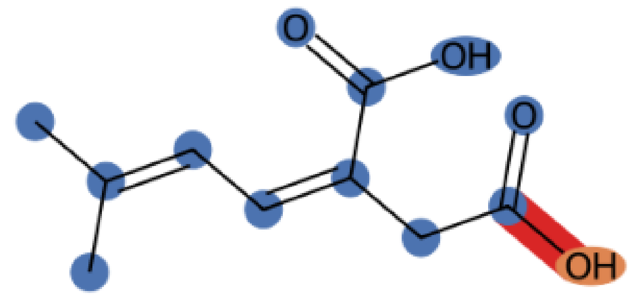


E · Example decompositions (ratio 0.3333)

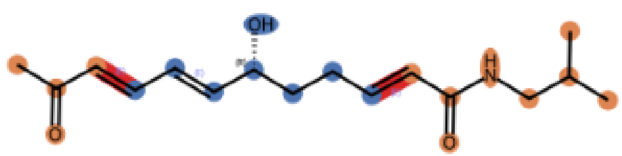
CNP0000074 k=1 core 20 atoms, R-groups [3]



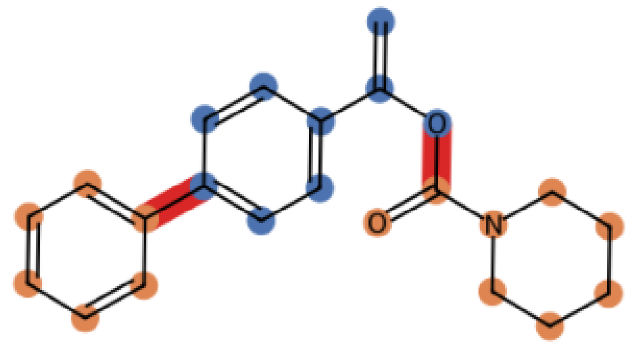
CNP0000161 k=1 core 12 atoms, R-groups [1]



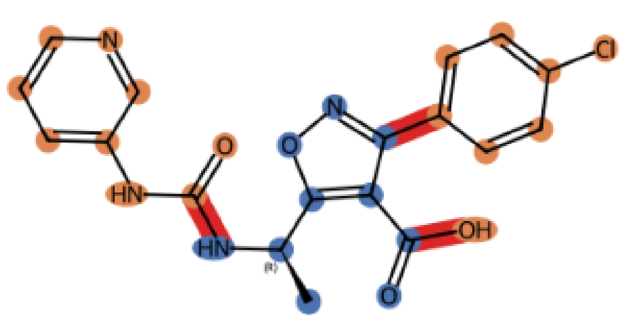
CNP0000307 k=2 core 8 atoms, R-groups [4, 8]



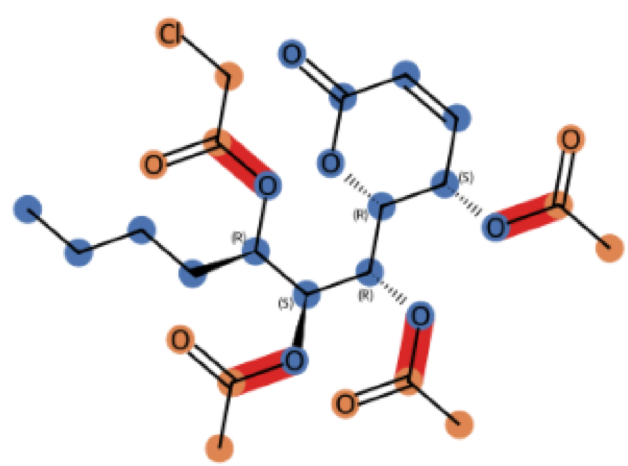
CNP0001292 k=2 core 9 atoms, R-groups [8, 6]



CNP0014826 k=3 core 10 atoms, R-groups [9, 7, 1]

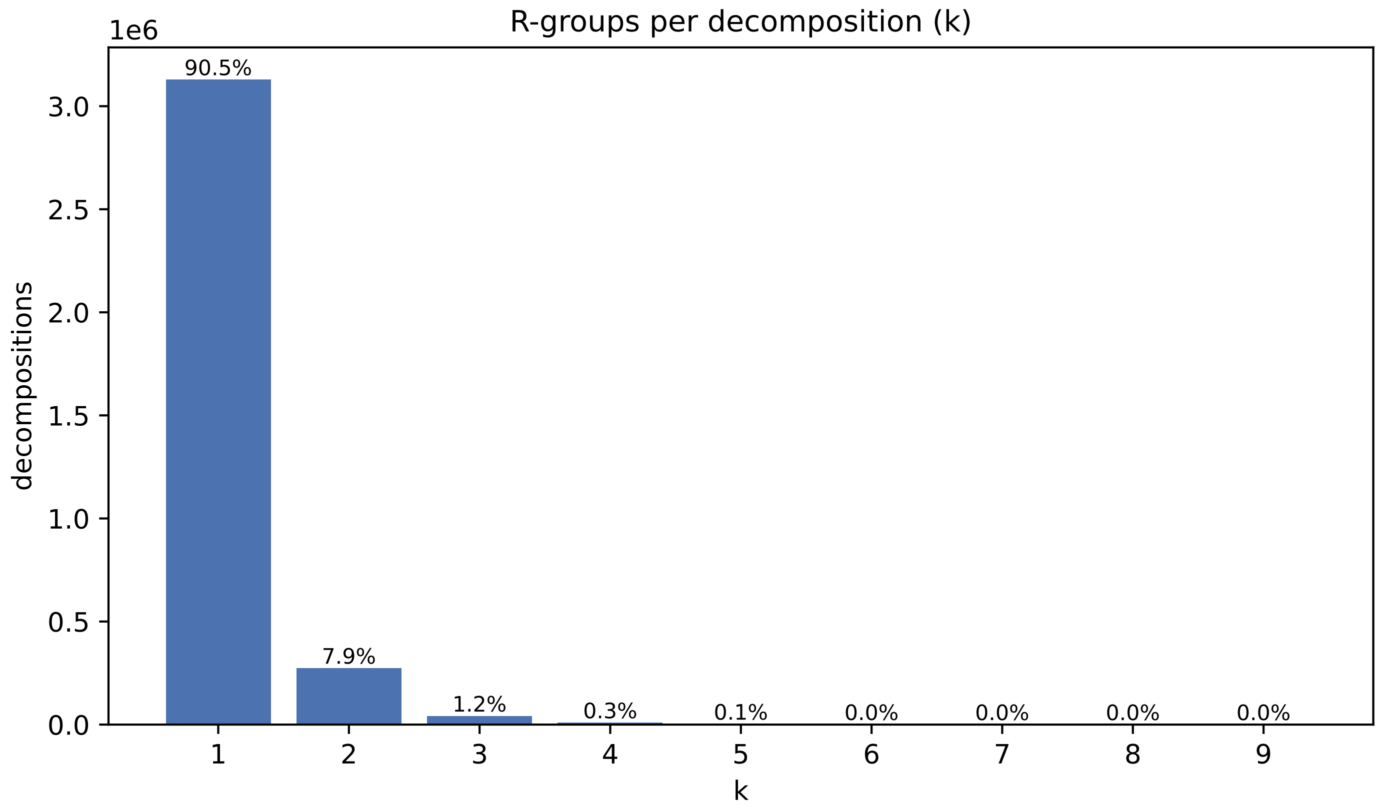


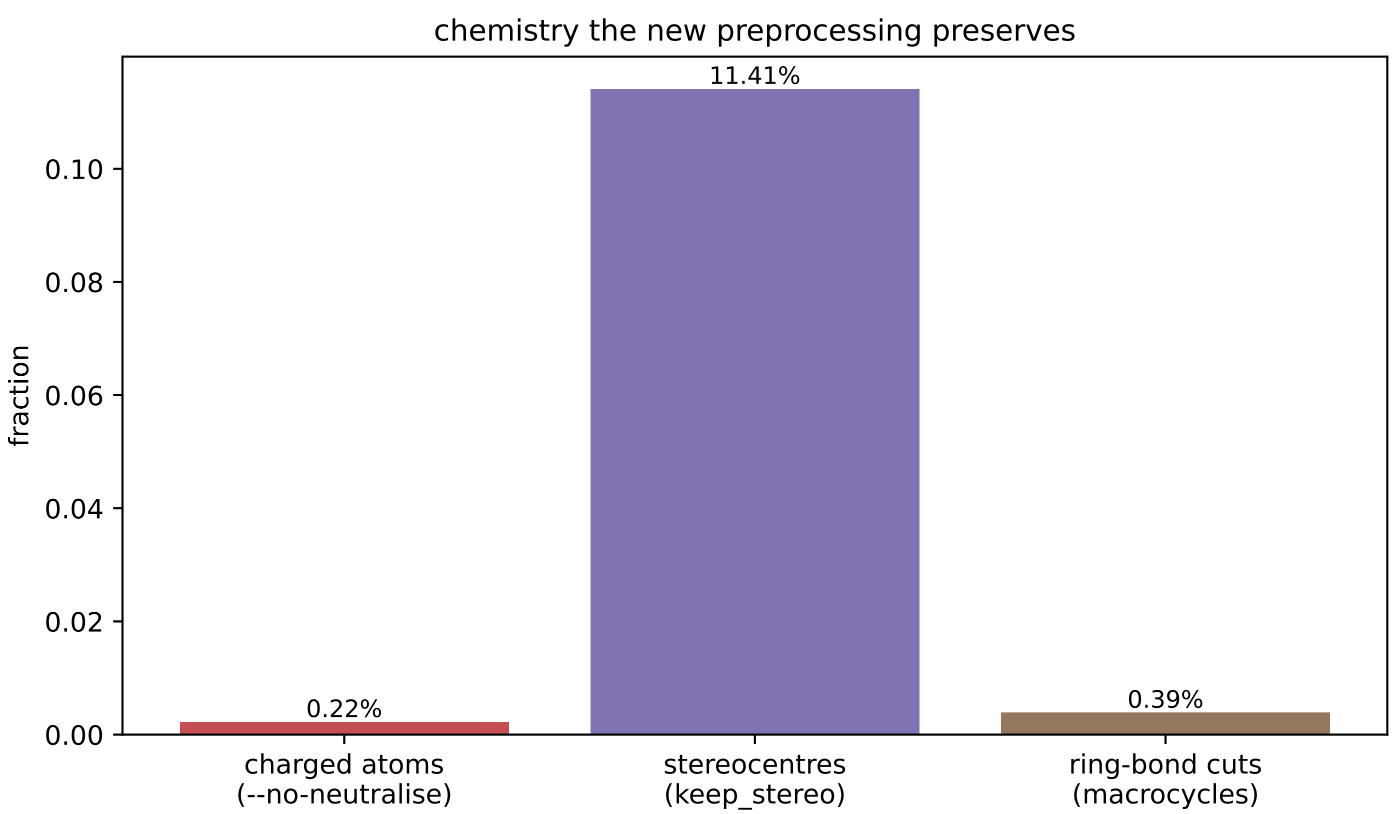
CNP0078333 k=4 core 18 atoms, R-groups [4, 3, 3, 3]



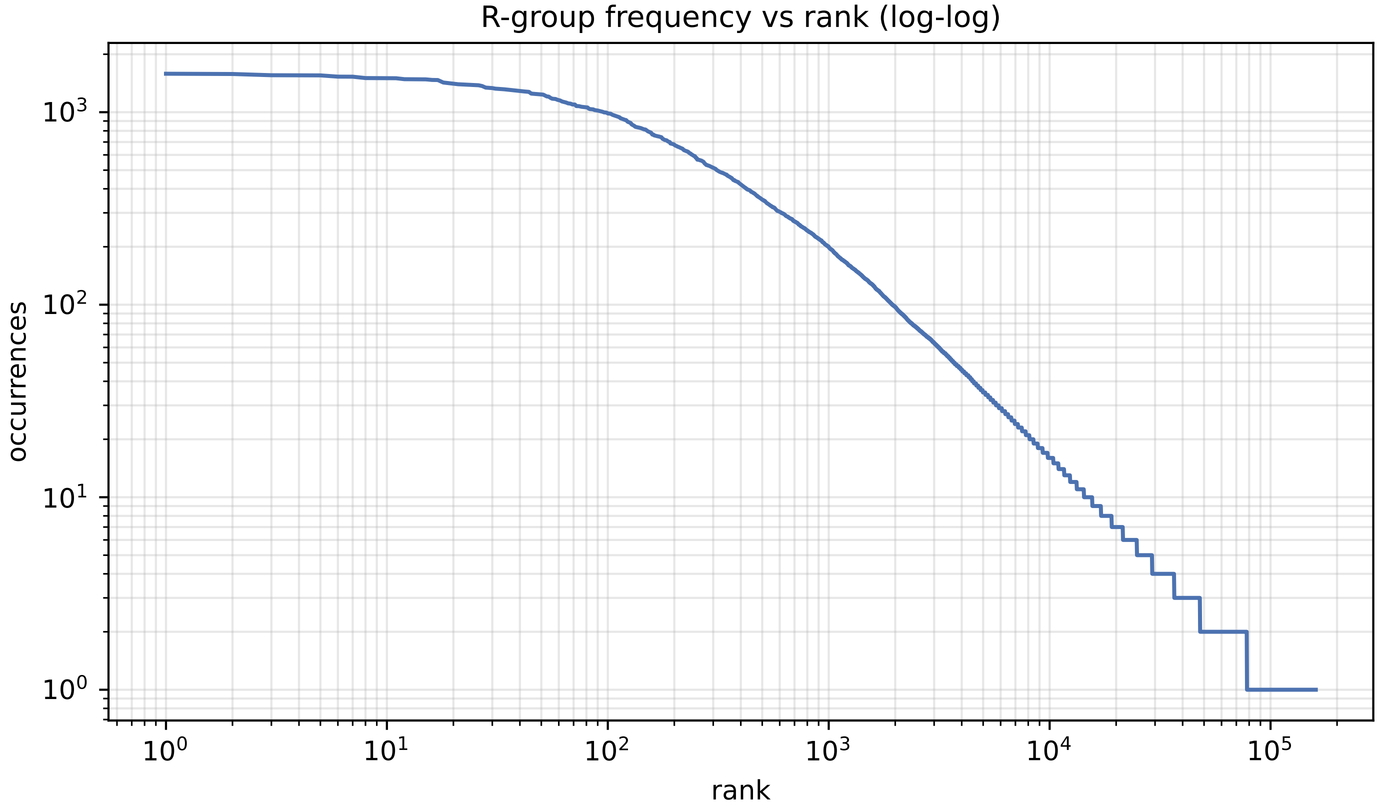
blue = core orange = R-groups red = cut bonds

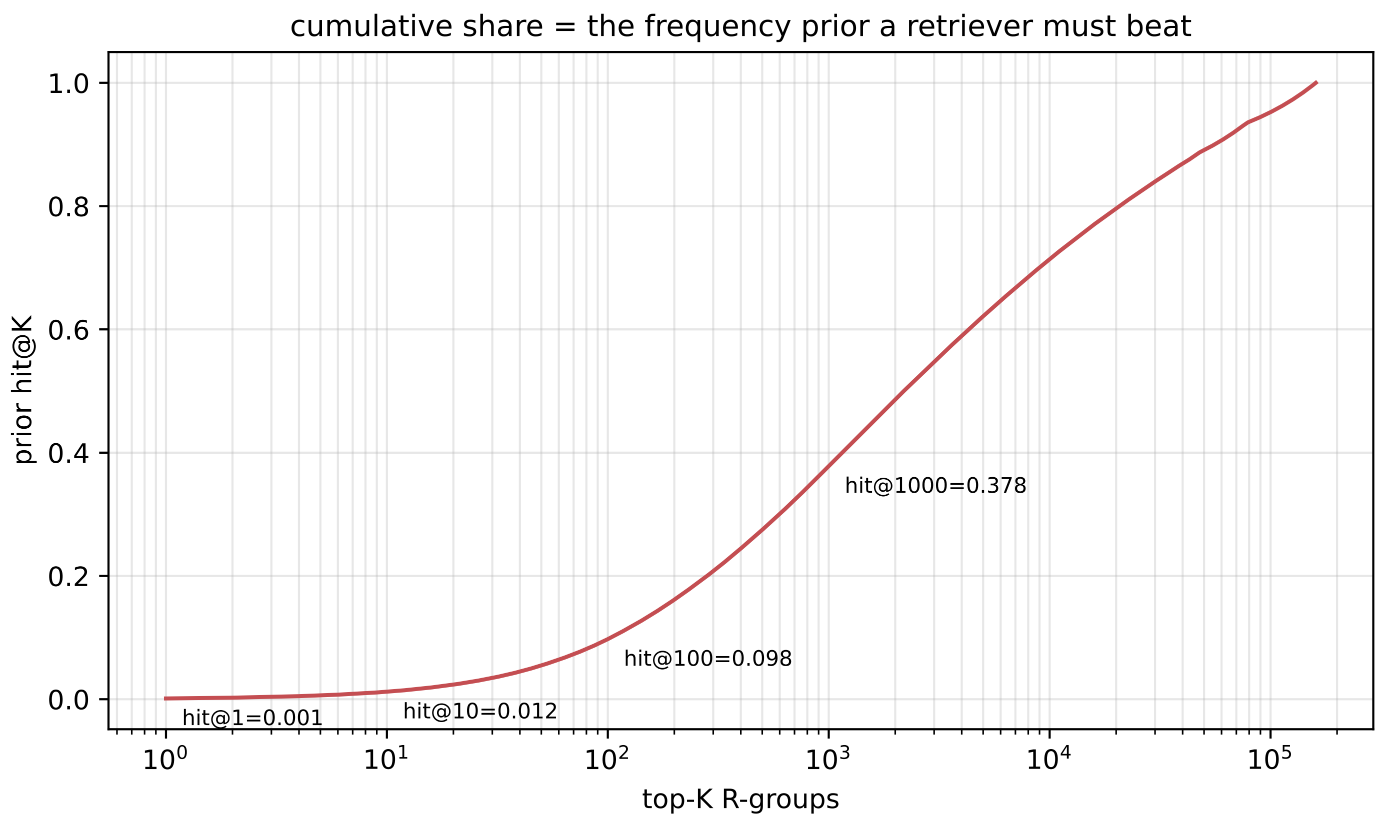
F · Decomposition shape & preserved chemistry





G · R-group library skew





H · R-group library

quantity	value
distinct R-groups	160,270
occurrences	1,266,473
effective size (exp H)	20,082
top-1 share	0.13%
top-10 cover	1.2%
top-100 cover	9.8%

Effective size, not the raw count, is the number of classes retrieval actually chooses between.

I · Most frequent R-groups

count	share	SMILES
1,584	0.13%	*C(=O)Cc1ccccc1
1,579	0.12%	*CCCO
1,556	0.12%	*C1CCCC1
1,555	0.12%	*NCc1ccccc1
1,553	0.12%	*C(=O)NC1CCCCC1
1,530	0.12%	*C#C
1,529	0.12%	*CC(N)=O
1,504	0.12%	*O[C@@H]1OC[C@H](O)[C@@H](O)[C@@H]1O
1,503	0.12%	*C(=O)OCc1ccccc1
1,501	0.12%	*OC(=O)C(C)C
1,501	0.12%	*Cc1ccccn1
1,484	0.12%	*CCCC(C)C
1,482	0.12%	*C(=O)c1cccs1
1,482	0.12%	*C(C)=CC
1,481	0.12%	*NC(C)C
1,471	0.12%	*CN(C)C
1,469	0.12%	*CCCC
1,426	0.11%	*C1OC[C@H](O)[C@@H](O)[C@@H]1O
1,414	0.11%	*Cc1ccc2c(c1)OCO2
1,400	0.11%	*N1C[C@@H]2C[C@H](C1)c1cccc(=O)n1C2